

Project brief: Ride-hailing trips

Group 4b · Data Science & Analytics Internship · 8 weeks

Your question. Which pickup zones generate the most revenue, and where are cancellations concentrated?

The situation

A ride-hailing operator covering fifteen zones in Accra wants to know where to concentrate drivers. Cancellations are a known problem but nobody has established whether they cluster in particular places or times.

Your audience is the operations lead. They are not an analyst. Whatever you build has to make sense to them without you standing next to it explaining.

Your data

Table	Rows	What is in it
trips	28,112	One row per trip. Date, driver, rider, pickup and dropoff zone, distance, duration, status, fare.
drivers	200	Name, vehicle type, join date, rating.
riders	1,000	Name, signup date, payment preference.
zones	15	Zone name, zone class.

trips is your fact table — one row is one trip. The other three are dimensions. There is no date table; you build that yourself in week 4.

The data has real problems in it. Missing values, duplicates, inconsistent spellings, dates in several formats, and some values that cannot be true. Finding them is week 2. Do not assume the data is clean because it loaded.

Connecting

Host	internship-db.coh86gwewtxb.us-east-1.rds.amazonaws.com
Port	5432
Database	internship
Username	group4b
Password	sent to your group lead separately
SSL	required
You read from	raw_rides (read-only)
You write to	group4b (yours alone)

You cannot write to raw_rides, and you cannot see group4a's schema. That is deliberate.

The eight weeks

Week	Phase	What you do
1	Scope	Meet as a group. Agree who leads which phase. Write down what you think 'done' looks like and commit it.

Week	Phase	What you do
2	Profile the data in SQL	Work through your query pack top to bottom. Find out what is in the data and what is wrong with it. Commit data_quality_notes.md.
3	Clean in Python	Fix what you found. Parse the dates, standardise the categories, decide what to do about duplicates, impossible values and orphan keys. Write cleaned tables back to your schema.
4	Build the model	Design your star schema in Power BI. One fact table, your dimensions, and a date table. Set the relationships. No visuals yet.
5	Measures and first draft	Write your DAX measures, then a rough dashboard. Ugly is expected at this stage.
6	Iterate	Fix what the draft exposed. Usually a missing dimension or a measure that does not slice correctly.
7	Freeze and rehearse	No new features. Formatting, documentation, and a timed run-through of your presentation.
8	Submit and present	Final submission and demo.

Scope freezes at the start of week 7. Nothing new gets added after that — polish only.

What you submit

- A Power BI file with a working star schema, not a flat table.
- A dashboard that answers the question above.
- A GitHub repository containing your SQL, your cleaning notebook, your data quality notes, and a README a stranger could follow.
- A ten-minute presentation.

How you are judged

	What we look at
Data handling	Did you find the problems, and can you justify what you did about each one?
The model	Is it a real star schema with working relationships and a date table?
The answer	Does the dashboard actually answer the question, or just show charts?
Clarity	Can the operations lead read it without you narrating?
Progress	Distance covered across the eight weeks, not only the final state.
Contribution	Individual commits. Everyone should have touched every tool.

Ground rules

- Bring something visible to every weekly call — a query, a chart, a thing that broke. Progress updates without an artifact do not count.
- Commit as you go, under your own name. Work that lives only on one laptop is work nobody can see.
- Rotate who leads each phase, so everyone touches every tool.
- Ask in the shared channel, not in private messages. Someone else has the same problem.
- Data quality findings are shared openly with group4a. Your analysis is not.

You are not expected to know how to do this yet. Most of the cohort is starting at beginner level, and the programme is built for that. What is expected is that you show up, ask early, and keep going when something breaks.